

Data Thinning and Risk Estimation

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Project Overview

In this project, we study how data thinning affects model estimation and risk evaluation in linear regression. Sample splitting is commonly used to avoid bias from reusing data, but it reduces statistical efficiency by discarding observations. Data thinning offers an alternative by creating two independent sets of the same data by adding constructed noise instead of reducing sample size. We analyze how the thinning parameter γ influences bias and variance in mean-squared error estimation.

In the Gaussian setting, thinning is defined as follows. Suppose

$$Y_i \sim \mathcal{N}(\mu_i, \sigma^2),$$

and let $W_i \sim \mathcal{N}(0, \sigma^2)$ independently. For $\gamma > 0$, define

$$Y_i^{(1)} = Y_i + \gamma W_i, \quad Y_i^{(2)} = Y_i - \frac{1}{\gamma} W_i.$$

Under the Gaussian setting, $Y^{(1)}$ and $Y^{(2)}$ are independent and share the same mean structure, but with inflated variances. This construction allows us to train a model on $Y^{(1)}$ and evaluate its risk using $Y^{(2)}$ without reusing the same noise realization.

Research Questions

We investigate the following questions:

1. How does data thinning affect the OLS estimator relative to the full-data estimator?
2. How does thinning influence the bias of empirical risk estimation?
3. How does the thinning parameter γ control this bias?
4. How should we tune γ to optimize model selection?

Theoretical Progress

We consider the linear model

$$Y = X\beta^* + \varepsilon, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Using the thinned outcomes $Y^{(1)}$, we define the data-thinned OLS estimator

$$\hat{\beta}_{DT} = (X^\top X)^{-1} X^\top Y^{(1)}.$$

We proved that the estimator remains unbiased:

$$\mathbb{E}[\hat{\beta}_{DT}] = \beta^*.$$

We also derived its variance:

$$\text{Var}(\hat{\beta}_{DT}) = (1 + \gamma^2)\sigma^2(X^\top X)^{-1}.$$

We are currently working on quantifying the bias of the data-thinned empirical risk estimator relative to the full-data risk.

Simulation Study

We conducted Monte Carlo simulations under the Gaussian linear model:

- $X \in \mathbb{R}^{n \times d}$ with $n = 100$, $d = 10$, and $X_{ij} \sim \mathcal{N}(0, 1)$,
- $\beta_i^* = 2^{-i+1}$,
- $Y = X\beta^* + \varepsilon$, where $\varepsilon_i \sim \mathcal{N}(0, 1)$.

Across 1000 repetitions, we sampled $W_i \sim \mathcal{N}(0, 1)$ and computed the bias

$$\hat{R}^{(2)}(\hat{\beta}^{(1)}) - R(\hat{\beta}),$$

where the first term evaluates empirical risk using $Y^{(2)}$, and the second term approximates the true population risk using an independent test sample. We plotted the average bias as a function of γ^2 to illustrate how thinning affects empirical risk estimation.

This project combines theoretical derivation and simulation to understand the tradeoff between independence and variance inflation introduced by thinning.